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Maximum Mean Discrepancy on Exponential Windows for Online Change Detection

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Abstract

Detecting changes is of fundamental importance when analyzing data streams and has many applications, e.g., in predictive maintenance, fraud detection, or medicine. A principled approach to detect changes is to compare the distributions of observations within the stream to each other via hypothesis testing. Maximum mean discrepancy (MMD), a (semi-)metric on the space of probability distributions, provides powerful non-parametric two-sample tests on kernel-enriched domains. In particular, MMD is able to detect any disparity between distributions under mild conditions. However, classical MMD estimators suffer from a quadratic runtime complexity, which renders their direct use for change detection in data streams impractical. In this article, we propose a new change detection algorithm, called Maximum Mean Discrepancy on Exponential Windows (MMDEW), that combines the benefits of MMD with an efficient computation based on exponential windows. We prove that MMDEW enjoys polylogarithmic runtime and logarithmic memory complexity and show empirically that it outperforms the state of the art on benchmark data streams.

1 Introduction

Data streams are possibly infinite sequences of observations that arrive over time. They can have different sources: sensors in industrial settings, online transactions from financial institutions, click monitoring on websites, online feeds, etc. Quickly detecting when a change takes place can yield useful insights, for example, about machine failure, malicious financial transactions, changes in customer preferences, and public opinions.

A *change* occurs if the underlying distribution of the data stream changes at a certain point in time. We call this moment *change point* (Gama, 2010); it is sometimes also referred to as *concept drift*. A principled and widely-used approach to detect changes is to use two-sample tests. The null hypothesis of such tests is that the data before and after the potential change point follow the same distribution. If the test rejects the hypothesis, one assumes that a change occurred.

One way to construct these tests is to use the kernel-based maximum mean discrepancy (MMD; Smola et al. 2007; Gretton et al. 2012), which one can interpret as a (semi-)metric on the space of probability distributions.[1] In the statistics literature, MMD is also known as energy distance (Székely & Rizzo, 2004;

[1] A function is a semimetric if it is a metric but can be zero for distinct elements.

2005); see Sejdinovic et al. (2013) for the equivalence. MMD relies on the kernel mean embedding (Berlinet & Thomas-Agnan, 2004, Ch. 4); it uses a kernel function to map a probability distribution to a reproducing kernel Hilbert space (RKHS; Aronszajn 1950) and quantifies the discrepancy of the two distributions as their distance in the RKHS. MMD is a metric if the kernel mean embedding is injective; the kernel is then called characteristic (Fukumizu et al., 2008; Sriperumbudur et al., 2010). When using a characteristic kernel, the MMD two-sample test allows to distinguish any distributions given that their kernel mean embeddings exist, which is guaranteed under mild conditions.

Two-sample tests based on MMD are widely applicable, as there exist kernel functions for a multitude of Euclidean and non-Euclidean domains, for example, strings (Watkins, 1999; Cuturi & Vert, 2005), graphs (Gärtner et al., 2003; Borgwardt et al., 2020), or time series (Cuturi, 2011; Király & Oberhauser, 2019). Another benefit of kernel-based two-sample tests is their high power. While, for Euclidean data, it has been shown that the power of such tests generally decreases in the high-dimensional setting (Ramdas et al., 2015), recent results (Cheng & Xie, 2024) establish that the power rather depends on the intrinsic dimensionality of the data. The intrinsic dimensionality is typically low in real-world settings so that the kernel-based two-sample tests there do not suffer the curse of dimensionality.

Despite these benefits, a well-known bottleneck of MMD-based approaches is their computational complexity. When comparing the distributions of two sets of data of sizes m and n, respectively, the computation of MMD with classical estimators is in $\mathcal{O}\left(m^2 + n^2\right)$, with a memory complexity in $\mathcal{O}\left(m + n\right)$. Naively computing MMD for each possible change point on a data stream with $t = m + n$ observations has a complexity in $\mathcal{O}\left(t^3\right)$ for each new observation. These properties render the direct application of MMD to change detection in data streams impractical.

In this paper, we introduce Maximum Mean Discrepancy on Exponential Windows (MMDEW), a change detection algorithm for data streams that solves the above bottleneck. Specifically, our **contributions** include the following.

- Our main contribution is MMDEW, a change detector based on an efficient online approximation of MMD. When considering the entire history of t observations, the proposed method has a memory requirement of $\mathcal{O}\left(\log t\right)$ and a runtime complexity of $\mathcal{O}\left(\log^2 t\right)$ for each new observation. Otherwise, the algorithm has constant runtime and memory requirements.
- To achieve these complexities, we introduce a new data structure, which allows to approximate the quadratic time MMD in an online setting. We accomplish the speedup by introducing windows that store summaries of the observations seen so far, and by storing a sample of logarithmic size of the observations per window.
- Our experiments on standard benchmark data sets show that MMDEW performs better than state-of-the-art change detectors on four out of the five tested data sets using the F_1-score. For the more challenging setting of short detection delays, the proposed algorithm is better on three out of six data sets.[2]

Outline. Section 2 summarizes related work. Section 3 introduces the definitions and Section 4 presents the proposed algorithm. We detail the experiments in Section 5. Section 6 concludes. We include illustrative proofs in the main text but defer technical proofs, additional details, and additional experiments to the appendices.

2 Related work

Change detection is an unsupervised task that has received and still is receiving a lot of interest. The earliest approaches, for example, Shewhart (1925); Page (1954), originated from quality control and require strong parametric assumptions on the pre and post-change distributions. More recent work in the parametric regime weakens these assumptions by allowing post-change distributions from a parametric family with an unknown parameter (Lorden, 1970; Siegmund & Venkatraman, 1995) or by allowing any post-change distribution (Sparks, 2000; Lorden & Pollak, 2005; Abbasi & Haq, 2019; Xie et al., 2023).

[2] Our code is available at `https://github.com/FlopsKa/mmdew-change-detector`.

In our setting, both the pre and post-change distribution are assumed to be unknown, which is a challenging setting that can be tackled with non-parametric approaches. We detail the approaches most related to our proposed method in the following and refer to Wang & Xie (2024) for a recent more extensive survey on parametric and non-parametric change detection methods.

A principled approach for comparing distributions in a stream in a non-parametric fashion is to use a corresponding statistical test. ADWIN (Bifet & Gavaldà, 2007) is a classic example but it is limited to univariate data and only detects changes in mean. ADWINK (Faithfull et al., 2019) alleviates the former by running one instance of ADWIN per feature and issues a change if a predefined number of the instances agree that a change occurred. Hence, the approach can only detect changes in the means of the marginal distributions and changes in higher moments or the covariance structure can not be detected. Still, the authors find that such an ensemble of univariate change detectors often outperforms multivariate detectors. WATCH (Faber et al., 2021) is a recent approach that uses a two-sample test based on the Wasserstein distance. However, the estimation of the Wasserstein distance requires density estimation, which is difficult for high-dimensional data (Scott, 1991). The method Dasu et al. (2009) is conceptually similar to our method, as it also relies on two-sample tests and is non-parametric, but it also requires density estimation.

In contrast, the computation of MMD-based two-sample tests does not become more difficult on high-dimensional data, which renders their usage for change detection on such data promising. We refer to Muandet et al. (2017) for a general overview of kernel mean embeddings and MMD.

There exist methods to compute MMD in the streaming setting, for example, linear time tests (Gretton et al., 2012), but their statistical power is low. Zaremba et al. (2013) introduce B-tests, which have higher power. However, both can not directly be used for change detection. Li et al. (2019) enable the estimation of MMD on data streams for change detection by introducing Scan B-statistics. Wei & Xie (2022) extend upon their work by considering multiple Scan B-statistics in parallel and introduce online kernel CUSUM. Another method enabling the computation of MMD on data streams is NEWMA (Keriven et al., 2020), which is based on random Fourier features (Rahimi & Recht, 2007; Sriperumbudur & Szabó, 2015), a well-known kernel approximation. NEWMA also allows detecting changes on streaming data. Harchaoui & Cappé (2007) apply kernel-based tests for offline change point detection on audio and brain-computer-interface data.

A conceptually different approach to find changes is using classifiers. D3 (Gözüaçik et al., 2019) maintains two consecutive sliding windows and trains a classifier to distinguish their elements. It reports a change if the classifier performance, measured by AUC, drops below a threshold. Another recent algorithm is IBDD (de Souza et al., 2021), which scales well with the number of features.

In our experiments in the main text, we compare MMDEW to ADWINK, WATCH, Scan B-Statistics, NEWMA, D3, and IBDD as these allow change detection on multivariate streams (in $\mathbb{R}^d$). These algorithms differ w.r.t. their runtime complexity, their theoretical properties, the data types that they can handle, and the types of changes that they can detect. We summarize their main properties in Table 1.[3] We consider the dimensionality d as constant for the complexities where its influence is dominated by other terms and for approaches not restricted to Euclidean domains. In Appendix D.1 and Appendix D.2, we collect additional experiments on synthetic data. In particular, we additionally compare the proposed method to online kernel CUSUM and to multivariate adaptations of the Cramer-von-Mises change point model (CvM CPM; Ross & Adams 2012) and non-parametric Focus (Romano et al., 2023).

3 Definitions and background

This section defines our problem and recalls kernels, the mean embedding, maximum mean discrepancy, and two-sample testing.

Problem definition. Let $(\mathcal{X}, \tau_{\mathcal{X}})$ be a topological space, $\mathcal{B}(\tau_{\mathcal{X}})$ the Borel sigma-algebra induced by $\tau_{\mathcal{X}}$, and $\mathcal{M}_1^+(\mathcal{X})$ the set of probability measures on $\mathcal{X}$ meant w.r.t. the measurable space $(\mathcal{X}, \mathcal{B}(\tau_{\mathcal{X}}))$. We consider a data stream, that is, a possibly infinite sequence of observations, $x_1, x_2, \ldots, x_t, \ldots$ for $t = 1, 2, \ldots$, and

[3] [a] We refer to their used implementation of the Wasserstein distance computation and the discussion therein (Mérigot, 2011, Ch. 6). [b] m is the number of random Fourier features and $m \ll d$. [c] The complexity results from the matrix inversion of the logistic regression model, which has cubic runtime cost. [d] Size of the constructed $q \times p$ image.